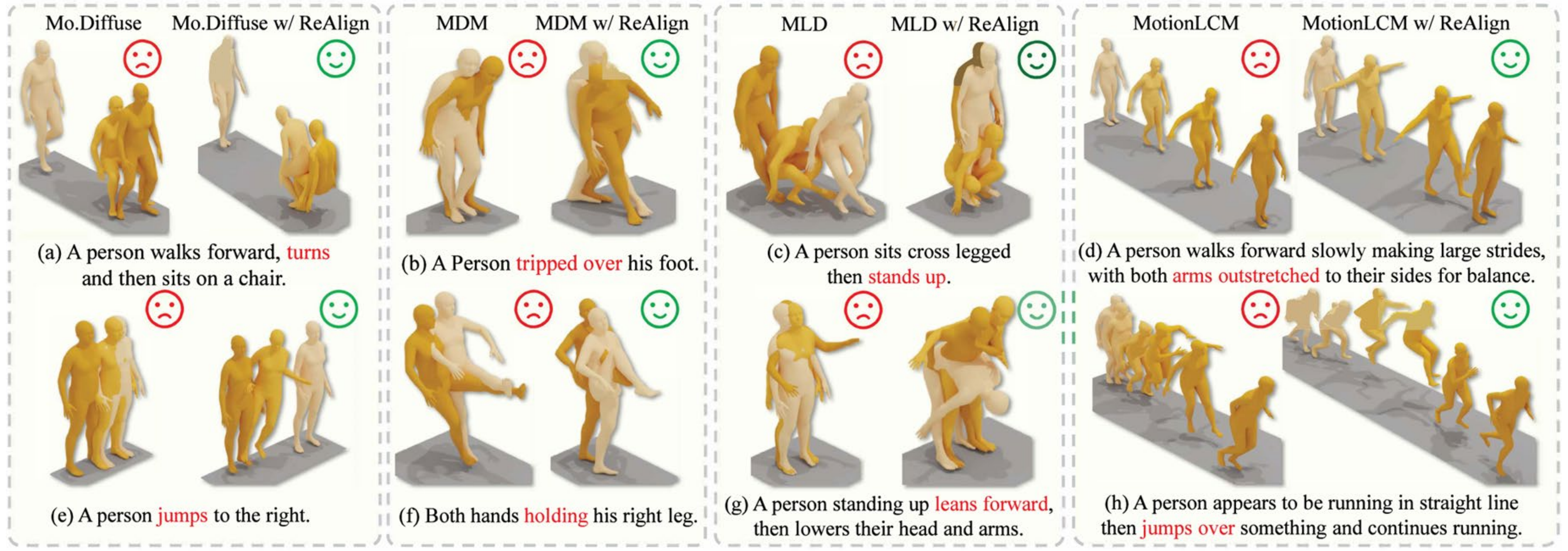
ReAlign: Text-to-Motion Generation via Step-Aware Reward-Guided Alignment

Association for the Advancement of Artificial Intelligence

Wanjiang Weng[1,2,*], Xiaofeng Tan[1,2,*], Junbo Wang[3], Guo-Sen Xie[4], Pan Zhou[5], Hongsong Wang[1,2,†]
[1]Southeast University, [2]PALM Lab, [3]Northwest Polytechnical University,
[4]Nanjing University of Science and Technology, [5]Singapore Management University

Introduction

Issue: Existing text-to-motion methods suffer from semantic misalignment and fail to rectify the denoising process.



(a) A person walks forward, turns and then sits on a chair. (b) A Person tripped over his foot. (c) A person sits cross legged then stands up. (d) A person walks forward slowly making large strides, with both arms outstretched to their sides for balance. (e) A person jumps to the right. (f) Both hands holding his right leg. (g) A person standing up leans forward, then lowers their head and arms. (h) A person appears to be running in straight line then jumps over something and continues running.

Key Observation: We observe that diffusion-based text to motion generation accumulates text-motion misalignment during denoising.

Question: Can we *improve text–motion alignment* and realism *at inference time?*

Contributions: We propose ReAlign, a plug and play reward guided sampling method with a step aware reward model to improve text motion alignment and motion realism during denoising without fine tuning the diffusion model.

Motivation: Diffusion based text to motion models can generate realistic motions but often drift from the text because sampling favors high probability regions and image trained text encoders do not capture motion dynamics. Since paired motion text data is limited, we guide the sampling process with a reward distribution learned from available pairs to improve alignment without fine tuning the diffusion model.

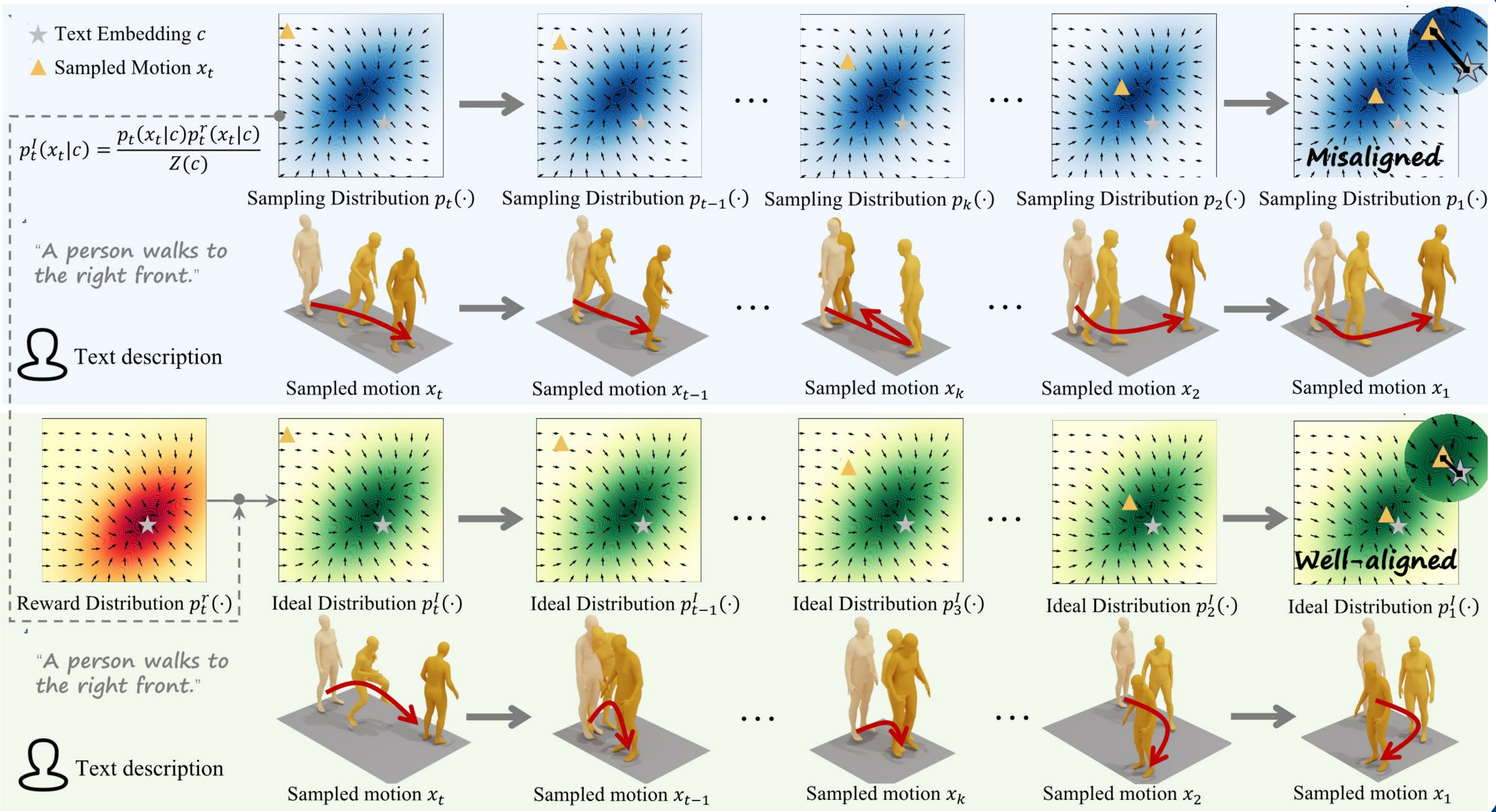
Contact

Email: wjweng@seu.edu.cn
txf0620@gmail.com

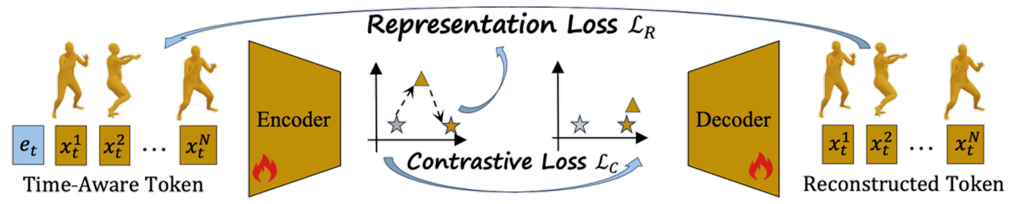
WeChat: wengwanjiang
txf_06_20

Author | Project Page | Code | Paper

Toy Example



Reward-Guided Sampling



Algorithm 1: Training Step-Aware Reward Model

Input: Step-aware reward model R_φ, training set $\mathcal{D}_{tr}$, timestep T range $[t_{\min}, t_{\max}]$, probability parameter ω, noise scheduler α.
Output: Step-aware reward model R_φ.

```
repeat
for (x, c) in D_tr do
    t ← 0                                        ▷ Initialize t
    if Uniform(0,1) > ω then
        t ← Uniform(t_min, t_max)                ▷ Add noise to motion
    end if
    x_t ~ N(√α_t x, (1 − α_t)I)                  ▷ Forward process
    L_RM(φ; x_t, c) ← L_C(φ; x_t, c) + L_R(φ; x_t, c)
                                                 ▷ Compute loss L_RM
    φ ← φ − ∇_φ L_RM(φ)                          ▷ Update parameter
end for
until converged
```

Reward Distribution. With both the step-aware reward model and the motion-to-motion reward, we define the dual-alignment reward as:

$$R(\mathbf{x}_t, c) = \mu R_\varphi(\mathbf{x}_t, c) + \eta R_m(\mathbf{x}_t, c), \tag{9}$$

where μ and η control the contributions of text-based and

Algorithm 2: Reward-Guided Denoise Process

Input: Diffusion model ϵ_θ, reward model R, training set $\mathcal{D}_{tr}$, condition c, timestep T.
Output: Generated motion $\mathbf{x}_0$.

```
x_T ~ N(0, I)
x^c = arg max_{x∈D_tr} R_φ(x, c)
for t = T, ··· , 1 do
    use x^c to obtain reward score
    ε ~ N(0, I) if t > 1 else ε = 0
    R(x_t, c) ← μR_φ(x_t, c) + ηR_m(x_t, c)
                                       ▷ Compute Reward R(x_t, c)
    x_{t−1} ← (1/√α_t)(x̄_{t−1} + √β_t ε) + ∇R(x_t, c).
                                       ▷ Denoise guided by reward model
end for
return x_0
```

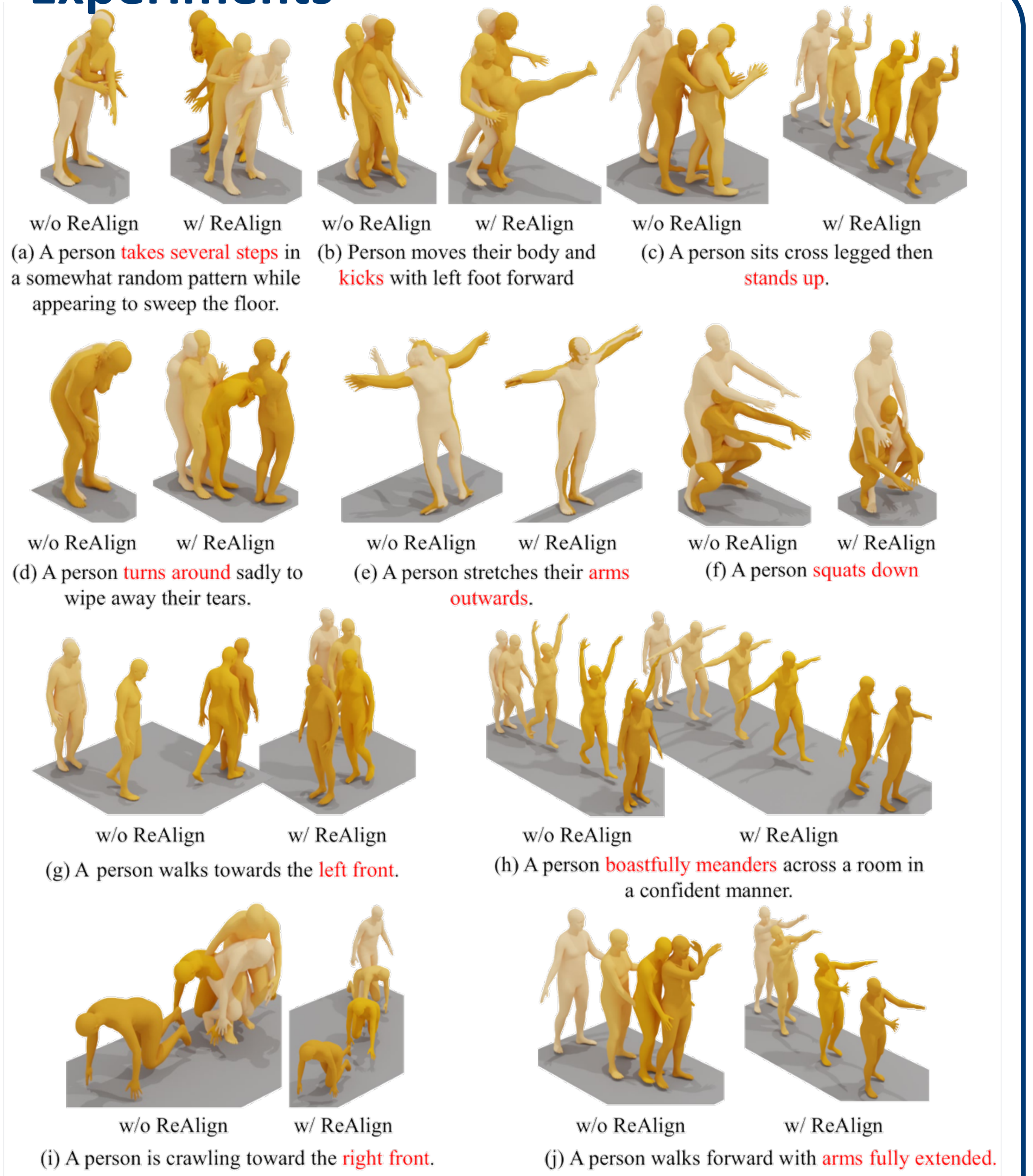
motion-based alignment. This reward formulation defines the reward distribution over noised motion as:

$$p_t^r(\mathbf{x}_t|c) = \exp\left(R(\mathbf{x}_t, c)\right) / Z^r(c). \tag{10}$$

Here, $Z^r(c) = \int \exp(R_\varphi(\mathbf{x}, c))\mathrm{d}\mathbf{x}$ is for normalization.

By integrating text-motion and motion-motion alignment, our approach constructs a robust reward signal that captures both semantic consistency and motion coherence. This enables more precise guidance of the diffusion sampling process, ensuring that generated motions are not only probable but also faithful to their textual descriptions.

Experiments



(a) A person takes several steps in a somewhat random pattern while appearing to sweep the floor. (b) Person moves their body and kicks with left foot forward. (c) A person sits cross legged then stands up. (d) A person turns around sadly to wipe away their tears. (e) A person stretches their arms outwards. (f) A person squats down. (g) A person walks towards the left front. (h) A person boastfully meanders across a room in a confident manner. (i) A person is crawling toward the right front. (j) A person walks forward with arms fully extended.

Method	R Precision ↑			FID ↓	MM Dist ↓	Diversity →
	Top 1	Top 2	Top 3			
Real	0.511	0.703	0.797	0.002	2.974	9.503
T2M (2022a)	$0.455^{\pm 0.002}$	$0.636^{\pm 0.003}$	$0.736^{\pm 0.003}$	$1.087^{\pm 0.002}$	$3.347^{\pm 0.008}$	$9.175^{\pm 0.002}$
MDM (2023)	$0.455^{\pm 0.006}$	$0.645^{\pm 0.007}$	$0.749^{\pm 0.006}$	$0.489^{\pm 0.047}$	$3.330^{\pm 0.25}$	$9.920^{\pm 0.083}$
T2M-GPT (2023a)	$0.492^{\pm 0.003}$	$0.679^{\pm 0.002}$	$0.775^{\pm 0.002}$	$0.141^{\pm 0.005}$	$3.121^{\pm 0.009}$	$9.722^{\pm 0.082}$
ReMoDiffuse (2023b)	$0.510^{\pm 0.005}$	$0.698^{\pm 0.006}$	$0.795^{\pm 0.004}$	$0.103^{\pm 0.004}$	$2.974^{\pm 0.016}$	$9.018^{\pm 0.75}$
Mo.Diffuse (2024)	$0.491^{\pm 0.001}$	$0.681^{\pm 0.001}$	$0.775^{\pm 0.001}$	$0.630^{\pm 0.001}$	$3.113^{\pm 0.001}$	$9.410^{\pm 0.049}$
OMG (2024)	-	-	$0.784^{\pm 0.002}$	$0.381^{\pm 0.008}$	-	$9.657^{\pm 0.085}$
MotionLCM (2025)	$0.502^{\pm 0.003}$	$0.698^{\pm 0.002}$	$0.798^{\pm 0.002}$	$0.304^{\pm 0.012}$	$3.012^{\pm 0.007}$	$9.607^{\pm 0.066}$
Mo.Mamba (2025b)	$0.502^{\pm 0.003}$	$0.693^{\pm 0.002}$	$0.792^{\pm 0.002}$	$0.281^{\pm 0.011}$	$3.060^{\pm 0.000}$	$9.871^{\pm 0.084}$
CoMo (2024)	$0.502^{\pm 0.002}$	$0.692^{\pm 0.007}$	$0.790^{\pm 0.002}$	$0.262^{\pm 0.004}$	$3.032^{\pm 0.015}$	$9.936^{\pm 0.066}$
ParCo (2025)	$0.515^{\pm 0.003}$	$0.706^{\pm 0.003}$	$0.801^{\pm 0.002}$	$0.109^{\pm 0.005}$	$2.927^{\pm 0.008}$	$9.576^{\pm 0.088}$
MARDM (2024)	$0.500^{\pm 0.004}$	$0.695^{\pm 0.003}$	$0.795^{\pm 0.003}$	$0.114^{\pm 0.007}$	-	-
MG-MotionLLM (2025b)	$0.516^{\pm 0.002}$	$0.706^{\pm 0.002}$	$0.802^{\pm 0.003}$	$0.303^{\pm 0.010}$	$2.952^{\pm 0.009}$	$9.960^{\pm 0.073}$
EnergyMoGen (2025)	$0.526^{\pm 0.003}$	$0.718^{\pm 0.003}$	$0.815^{\pm 0.002}$	$0.176^{\pm 0.006}$	$2.931^{\pm 0.007}$	$\mathbf{9.500}^{\pm 0.091}$
MLD (2023b)	$0.481^{\pm 0.003}$	$0.673^{\pm 0.003}$	$0.772^{\pm 0.002}$	$0.473^{\pm 0.013}$	$3.196^{\pm 0.010}$	$9.724^{\pm 0.082}$
w/ ReAlign (Ours)	$0.567^{\pm 0.003}$ **(+17.9%)**	$0.759^{\pm 0.003}$ **(+12.8%)**	$0.848^{\pm 0.003}$ **(+9.8%)**	$0.195^{\pm 0.005}$**(+58.8%)**	$2.704^{\pm 0.007}$ **(+15.4%)**	$9.474^{\pm 0.068}$ **(+86.9%)**
MLD++(2025)	$0.548^{\pm 0.003}$	$0.738^{\pm 0.003}$	$0.829^{\pm 0.002}$	$0.073^{\pm 0.003}$	$2.810^{\pm 0.008}$	$9.658^{\pm 0.089}$
w/ ReAlign (Ours)	$\mathbf{0.572}^{\pm 0.002}$ (+4.4%)	$\mathbf{0.764}^{\pm 0.002}$ (+3.5%)	$\mathbf{0.852}^{\pm 0.001}$ (+2.8%)	$\mathbf{0.055}^{\pm 0.003}$ (+24.7%)	$\mathbf{2.648}^{\pm 0.008}$ (+5.8%)	$9.478^{\pm 0.055}$ **(+83.9%)**

	Methods	Noise	Text-Motion Retrieval↑					Motion-Text Retrieval↑				
			R@1	R@2	R@3	R@5	R@10	R@1	R@2	R@3	R@5	R@10
HumanML3D	TEMOS (2022)	✗	40.49	53.52	61.14	70.96	84.15	39.96	53.49	61.79	72.40	85.89
	T2M (2022b)	✗	52.48	71.05	80.65	89.66	**96.58**	52.00	71.21	81.11	89.87	<u>96.78</u>
	TMR (2023)	✗	67.16	81.32	86.81	91.43	95.36	67.97	81.20	86.35	91.70	95.27
	LaMP (2025)	✗	67.18	81.90	87.04	**92.00**	95.73	68.02	82.10	87.50	92.20	**96.90**
	ReAlign (ours)	✓	**67.59**	**82.24**	**87.44**	<u>91.97</u>	<u>96.28</u>	**68.94**	**82.86**	**87.95**	**92.44**	<u>96.28</u>
KIT-ML	TEMOS (2022)	✗	43.88	58.25	67.00	74.00	84.75	41.88	55.88	65.62	75.25	85.75
	T2M (2022b)	✗	42.25	62.62	75.12	87.50	96.12	39.75	62.75	73.62	86.88	95.88
	TMR (2023)	✗	49.25	69.75	78.25	87.88	95.00	50.12	67.12	76.88	88.88	94.75
	ReAlign (ours)	✓	**52.84**	**71.66**	**82.96**	**91.19**	**97.59**	**52.98**	**72.87**	**84.38**	**92.61**	**96.87**